# Object-Oriented Analysis and Design

**DEPT X471.91**

3 Units

Fall 2014

## Class Meeting Information

- Start date: September 29, 2014
- End date: December 07, 2014
- Modality: Online

## Instructor Information

Name: Amar Kota
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Amar Kota, M.S., is a Development Manager at the computer software firm DNTG in Buena Park, CA. He has over 15 years of IT experience including teaching various IT courses at Fortune 1000 companies, private institutions and community colleges across Southern California. He specializes in web application and relational database architecture and development using J2EE and Microsoft .NET technologies.

## Course Description

Learn how to develop better, more complex software by applying good analysis and design methodologies. You'll learn how to approach software development systematically and how standardized notation such as the Unified Modeling Language (UML) allows you to map out the functionality of an application before writing any code. You'll also learn good practices for defining classes and methods so that your applications run efficiently and are easier to maintain. Anyone using object-oriented programming language including C#, Java, VB.NET, or C++ can benefit from this course.

## Prerequisites — Classes or Knowledge Required Before Taking This Course

- Course work or experience using an object-oriented programming language such as Java, C# or VB.NET.
- A UML tool that allows you to export diagrams as an image (ex. JPG / GIF) or print as a PDF.

## Course Sequencing

- Introduction to UML (I&C SCI X471.71) is recommended.

 

**Course Objectives**

***At the end of this course, students will be able to:***

- Understand the basic concepts of object-oriented analysis and design (such as inheritance, aggregation, polymorphism, etc.)
- Students will have comprehensive understanding of the Unified Modeling Language.
- Apply the object-oriented principles and methods to solve practical, real-world problems.
- Apply the UML to various phases of software development, such as requirement analysis, system design, object design and more.

**Course Material**

- Applying UML and Patterns: Introduction to OOA/D & Iterative Development, Third Edition, Larman, Prentice Hall
- UML Distilled: A Brief Guide to the Standard Object Modeling Language, Third Edition, Fowler et al, Addison-Wesley

**Course Outline**

## Week 1

**Introduction to OOAD**
By the end of this lesson, students will be able to :

- Know the scope of OO analysis and design
- Define an object
- Understand the object model

**Method(s) of Instruction:**

- Threaded discussion forums
- Lesson presentation

**Assignments Due:**

- OOAD Intro Quiz

## Week 2

**Requirements Analysis**
By the end of this lesson, students will be able to :

- Analyze business requirements

- Identify quality attributes
- Elicit requirements

**Method(s) of Instruction:**

- Threaded discussion forums
- Lesson presentation
- Book readings

**Assignments Due:**

- None

## Week 3

**Use Cases**
By the end of this lesson, students will be able to :

- Understand the POST system
- Create use cases

**Method(s) of Instruction:**

- Threaded discussion forums
- Lesson presentation
- Book readings
- Case Study

**Assignments Due:**

- Use Case Diagrams

## Week 4

**Conceptual Model**
By the end of this lesson, students will be able to :

- Identify potential classes from use cases
- Create a conceptual model

**Method(s) of Instruction:**

- Threaded discussion forums
- Lesson presentation
- Book readings
- Case Study

**Assignments Due:**

- Conceptual Model Diagram

## Week 5

**System Behavior**
By the end of this lesson, students will be able to :

- Create sequence diagrams
- Create interaction diagrams
- Create state diagrams

**Method(s) of Instruction:**

- Threaded discussion forums
- Lesson presentation
- Book readings
- Case Study

**Assignments Due:**

- System Behavior Diagrams

## Week 6

**GRASP**
By the end of this lesson, students will be able to :

- Know and apply the nine components of GRASP

**Method(s) of Instruction:**

- Threaded discussion forums
- Lesson presentation
- Book readings

**Assignments Due:**

- None

## Week 7

**Visibility**
By the end of this lesson, students will be able to :
- Determine visibility
- Understand the Law of Demeter
- Create robust class diagrams

**Method(s) of Instruction:**

- Threaded discussion forums
- Lesson presentation
- Book readings
- Whitepaper

**Assignments Due:**

- Paperboy / Wallet essay

## Week 8

**Design Patterns**
By the end of this lesson, students will be able to :
- Learn about the different design patterns that can be used with OO languages

**Method(s) of Instruction:**

- Threaded discussion forums
- Lesson presentation
- Book readings

**Assignments Due:**

- Design Pattern Identification Quiz

## Week 9

**OO Metrics**
By the end of this lesson, students will be able to :
- Read and define various metrics used in OOAD
- Calculate metrics in an OO application

**Method(s) of Instruction:**

- Threaded discussion forums
- Whitepaper

**Assignments Due:**

- None

## Week 10

**OOAD Conclusion**
By the end of this lesson, students will be able to :
- Have a thorough understanding of OOAD

**Method(s) of Instruction:**

- Wiki

**Assignments Due:**

- None

## Evaluation and Grading

***Evaluation of Student Performance Weighted as Percentages of the Total Grade***

| | |
|---|---|
| Individual Assignments (6 Total) | 80% |
| Participation in class discussions | 20% |
| | 100% |

***Grading Scale***

A = 90% – 100%
B = 80% – 89%
C = 70% – 79%
D = 60% – 69%
F = 59% or less

To give the maximum amount of flexibility for working students, all individual assignments are due by the course's end date. Assignments submitted after the course's end date will not be accepted.

## Code of Conduct

All participants in the course are bound by the University of California Code of Conduct, found at http://www.ucop.edu/ethics-compliance-audit-services/_files/stmt-stds-ethics.pdf

## Netiquette

In an online course, the majority of our communication takes place in the course forums. However, when we have a need for communication that is private, whether personal, interpersonal, or professional, we will use individual email or telephone. Our primary means of communication is written. The written language has many advantages: more opportunity for reasoned thought, more ability to go in-depth, and more time to think through an issue before posting a comment. However, written communication also has certain disadvantages, such a lack of the face-to-face signaling that occurs through body language, intonation, pausing, facial expressions, and gestures. As a result, please be aware of the possibility of miscommunication and compose your comments in a positive, supportive, and constructive manner.

## Academic Honesty Policy

The University is an institution of learning, research, and scholarship predicated on the existence of an environment of honesty and integrity. As members of the academic community, faculty, students, and administrative officials share responsibility for maintaining this environment. It is essential that all members of the academic community subscribe to the ideal of academic honesty and integrity and accept individual responsibility for their work. Academic dishonesty is unacceptable and will not be tolerated at the University of California, Irvine. Cheating, forgery, dishonest conduct, plagiarism, and collusion in dishonest activities erode the University's educational, research, and social roles.

Students who knowingly or intentionally conduct or help another student engage in dishonest conduct, acts of cheating, or plagiarism will be subject to disciplinary action at the discretion of UC Irvine Extension.

**Disability Services**

If you need support or assistance because of a disability, you may be eligible for accommodations or services through the Disability Service Center at UC Irvine. Please contact the DSC directly at (949) 824-7494 or TDD (949) 824-6272. You can also visit the DSC's website: http://www.disability.uci.edu/. The DSC will work with your instructor to make any necessary accommodations. Please note that it is your responsibility to initiate this process with the DSC.